

Algebra First-Year Exam, January 2026, Part 2

Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. Prove that the following polynomials are irreducible in $\mathbb{Q}[x]$, stating clearly any theorems you wish to apply.

(a) $x^4 - 2x^3 + 6x^2 - x + 5$

(b) $x^5 - 4x + 2$

(c) $x^4 + x^3 + x^2 + x + 1$

2. Let R be a commutative ring with $1 \neq 0$.

- (a) Define what it means for an element $r \in R$ to be nilpotent.
- (b) Prove that the set of all nilpotent elements of R forms an ideal (called the nilradical of R).
- (c) Show that the nilradical is contained in every prime ideal of R .

3. Let $R = \mathbb{Z}[\sqrt{-3}] = \{a + b\sqrt{-3} \mid a, b \in \mathbb{Z}\}$.

- (a) Show that the function $N(a + b\sqrt{-3}) = a^2 + 3b^2$ is multiplicative, that is, $N(rs) = N(r)N(s)$ for all $r, s \in R$.
- (b) Prove that R is not a unique factorization domain.
- (c) Is R a principal ideal domain? Justify your answer.

4. Let R be a commutative ring with 1 and let M be a module over R .

- (a) State what it means for M to be a Noetherian R -module, and for R to be a Noetherian ring.
- (b) Prove that if M is a Noetherian R -module, then every nonempty set of submodules has a maximal element.
- (c) Give an example of a non-Noetherian commutative ring with 1.

5. Using the rational canonical form, find one representative of each conjugacy class of elements of order 3 in $\text{GL}(4, \mathbb{Q})$.

6. Let $K \subseteq E$ be fields.

- (a) Define what it means for an element of E to be algebraic over K .
- (b) Prove that if $\alpha, \beta \in E$ are algebraic over K , then $K(\alpha, \beta)$ is finite-dimensional over K .
- (c) Deduce that the elements of E that are algebraic over K form a field.